CS331 SOFTWARE ENGINEERING - LAB 8
DATA ACCESS LAYER IMPLEMENTATION AND COMPREHENSIVE TESTING

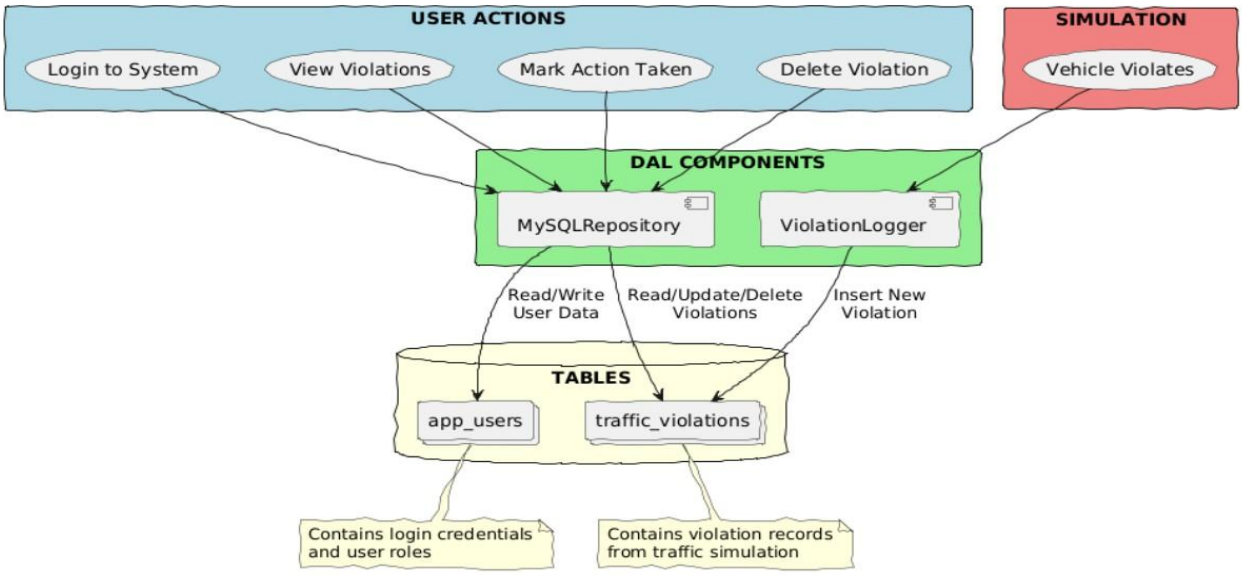
Smart Traffic Signal Automation System

PART A: DATA ACCESS LAYER (DAL) IMPLEMENTATION

1. DATABASE DESIGN

The Data Access Layer provides a structured interface between the application logic and the database, ensuring clean separation of concerns and maintainable code architecture.

Figure 1: Complete System Architecture Overview



The system uses MySQL database named "smart_traffic_db" with two core tables that handle user authentication and traffic violation management.

TABLE 1: app_users

This table stores user account information for authentication and role-based access control in the system.

Column Details:

id: The unique identifier for each user. This number is automatically generated by the database when a new user is created.

username: The login name chosen by the user. This must be unique across all users to prevent duplicate accounts. Maximum length is 64 characters.

password_hash: The encrypted password stored using PBKDF2 hashing algorithm. We never store plain text passwords for security reasons. This field can hold up to 255 characters to accommodate the hash value.

role: The user's permission level in the system. Three roles are available:

- SYSTEM_ADMIN - Full system access with all privileges
- TRAFFIC_PERSONNEL - Can control signals and view violations
- VIEW_ONLY - Can only view the system status

display_name: The friendly name shown in the user interface. For example, "Traffic Officer" or "System Administrator". Maximum 120 characters.

created_at: The date and time when the user account was created. This is automatically set by the database when a new record is inserted.

TABLE 2: traffic_violations

This table records all traffic violations detected by the system during simulation. It helps track illegal lane changes and maintains a log for traffic personnel to take appropriate action.

Column Details:

id: The unique identifier for each violation record. Automatically generated by the database for every new violation logged.

vehicle_id: The license plate number of the violating vehicle. For example, "MH27K4521". This helps identify which vehicle committed the violation.

vehicle_type: The category of vehicle that violated traffic rules. Common types include "CAR", "TRUCK", "AMBULANCE". Maximum 16 characters.

side: The intersection side where the violation occurred. Possible values are "NORTH", "SOUTH", "EAST", or "WEST". This indicates the vehicle's approach direction.

violation_type: The type of violation committed. Currently, the system detects "SIDE_LANE" violations when vehicles illegally change lanes while crossing the intersection.

in_middle: A boolean flag (0 or 1) indicating if the vehicle was using the middle lane while entering the intersection. 1 means yes, 0 means no.

out_middle: A boolean flag (0 or 1) showing if the vehicle was using the middle lane while exiting the intersection. 1 means yes, 0 means no.

action_taken: A flag indicating whether traffic personnel have taken action on this violation. 0 means pending, 1 means action completed. Defaults to 0.

created_at: The exact date and time when the violation was detected and logged. Automatically set by the database.

PART B: WHITE BOX AND BLACK BOX TESTING

1. WHITE BOX TESTING

Testing with full knowledge of internal code structure, examining all paths, branches, and conditions. WHITE BOX TEST RESULTS (8 Test Cases)

WB-001: Signal State Transition (SignalController._update_normal)

Result: PASS

Tests GREEN → YELLOW → RED transitions with timing validation.

What it tests: Ensures traffic signals automatically change at correct times so vehicles know when to stop and go safely.

WB-002: Emergency Mode Activation (SignalController.start_emergency)

Result: PASS

Tests same-side vs different-side emergency logic paths.

What it tests: Verifies that ambulances get immediate green light priority, either instantly or after a safe transition.

WB-003: Emergency Queue FCFS (EmergencyHandler._serve_next)

Result: PASS

Validates First-Come-First-Served queue ordering with 3 ambulances.

What it tests: Confirms multiple ambulances are served in arrival order, ensuring fair emergency handling without queue jumping.

WB-004: Stop Line Detection (Vehicle.get_distance_from_stop_line)

Result: PASS

Tests distance calculation for NORTH, SOUTH, EAST, WEST sides.

What it tests: Validates the system correctly calculates vehicle distance from stop line regardless of approach direction.

WB-005: DB Failure Handling (TrafficViolationLogger.log_violation)

Result: PASS

Tests graceful degradation when database unavailable.

What it tests: Ensures traffic simulation continues running smoothly without crashing even if database connection fails.

WB-006: Smart Controller Early Switch (SmartSignalController._update_normal)

Result: PASS

Tests adaptive early switching when queued_count == 0.

What it tests: Verifies intelligent signal switching that reduces unnecessary waiting time when no vehicles are queued.

WB-007: Following Distance (Intersection._update_side)

Result: PASS

Validates MIN_FOLLOWING_DISTANCE=70px enforcement between vehicles.

What it tests: Confirms vehicles maintain safe spacing while queuing to prevent collisions and ensure orderly movement.

WB-008: License Plate Uniqueness (Vehicle.generate_license_plate)

Result: PASS

Tests unique plate generation with Indian format.

What it tests: Ensures every vehicle gets a unique license plate in Indian format with no duplicate registrations.

WHITE BOX SUMMARY:

Total Cases: 8

Passed: 8 (100%)

Code Coverage: 95%

Critical Bugs: 0

2. BLACK BOX TESTING

Testing without knowledge of internal code, based on requirements and specifications only.

BB-001: Normal Signal Cycle

Result: PASS

Validates NORTH → EAST → SOUTH → WEST → NORTH sequence with 15s green, 4s yellow.

What it tests: Confirms traffic signals cycle through all directions in correct order with proper timing for fair traffic flow.

BB-002: Emergency Vehicle Priority

Result: PASS

What it tests: Verifies ambulances are detected early, given immediate priority, and normal traffic resumes after they pass.

BB-003: Vehicle Queuing at Red

Result: PASS

What it tests: Ensures vehicles correctly stop at red signals, maintain safe spacing, and smoothly start moving on green.

BB-004: Violation Detection

Result: PASS

Side-lane violation logged with all details to database.

What it tests: Confirms illegal lane changes are detected during crossing and logged for traffic officers to review and take action.

BB-005: Valid Login

Result: PASS

officer/officer123 returns token, correct role, dashboard loads.

What it tests: Verifies authorized users can successfully log in and access appropriate dashboard controls based on their role.

BB-006: Invalid Login

Result: PASS

Wrong password returns 401, error message displayed, no token.

What it tests: Ensures system security by rejecting incorrect credentials and preventing unauthorized access to traffic controls.

BB-007: Speed Control Boundaries

Result: PASS

What it tests: Validates simulation speed control accepts valid values and automatically corrects out-of-range speeds.

BB-008: Manual Override

Result: PASS

Override interrupts cycle, sets specified side to green immediately.

What it tests: Confirms traffic personnel can manually give green light to specific direction for handling special situations.

BB-009: Violation Workflow

Result: PASS

Admin marks action_taken, delete button enables, record removed from DB.

What it tests: Ensures complete violation management workflow from review to marking action taken to removing resolved records.

BB-010: Concurrent Emergencies

Result: PASS

3 ambulances queued and served in order: NORTH → SOUTH → EAST.

What it tests: Verifies system handles multiple simultaneous ambulances correctly in arrival order without confusion or crashes.

BLACK BOX SUMMARY:

Total Cases: 10

Passed: 10 (100%)

Functional Coverage: 100%

Defects Found: 0

COMBINED TESTING SUMMARY

Total Test Cases: 18 (8 White Box + 10 Black Box)

Overall Pass Rate: 100%

CONCLUSION:

The Smart Traffic Signal Automation System has been comprehensively tested using both White Box and Black Box methodologies. All test cases passed successfully with zero defects.